

Partially coinciding exponents: $d = 3$, $h = (0, 0, 1)$

$$Z(n) \sim \frac{b S_{1/2}(a)}{4} n^{-1/2} \log n \text{ (multiplicity two, not three)}$$

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2026-09-06

Abstract

The multiplicity in `thm:TaylorTree` counts the coordinates attaining the *minimal* candidate exponent. Units 37 and 39 covered the two extremes (all exponents distinct: no logarithm; all equal: $(\log n)^{d-1}$). This unit does the first mixed case, $k = (1, 1, 1)$, $h = (0, 0, 1)$ with exponents $(\frac{1}{2}, \frac{1}{2}, 1)$: the exact chart formula $Z(n) = (b/\sqrt{n}) (H(L) - K(L))$, $L = \sqrt{n} b^3$, with H the logarithmic primitive and $K(L) = \int_0^L x^{-2} F_1 = F_0(L) - F_1(L)/L$ bounded by A , hence $Z(n) \sim (b S_{1/2}(a)/4) n^{-1/2} \log n$. The non-minimal coordinate contributes only the bounded K and the factor b . Zero `sorry/axiom`.

1 Setting

Integrating the innermost weighted coordinate gives $(\sqrt{n} uv)^{-2} F_1(\sqrt{n} uvb)$ (unit 37's inner identity). The v -integral is the noncritical substitution $\int_0^b v^{-2} F_1(cv) dv = c K(cb)$, and the outer u -integral is the critical one, $\int_0^b K(cu)/u du = \int_0^{cb} K(x)/x dx$. The last integral is evaluated in closed form: since $K(x)/x = F_0(x)/x - x^{-2} F_1(x)$ on $x > 0$, it equals $H(L) - K(L)$.

2 The things formalised

```
theorem abs_quadMomentPrimitive_le ( a t : ) (h : 0 < ) :
  |quadMomentPrimitive a t| Real.exp ( * a ^ 2 / 2 ) * t ^ 2
theorem quadPowerPrimitive_continuous ( a : ) (h : 0 < ) :
  Continuous (quadPowerPrimitive a)
theorem quadPowerPrimitive_le_mass ... : quadPowerPrimitive a L quadMass a
theorem integral_quadMomentPrimitive_div_sq_eq ( a c b : ) (hc : 0 < c) (hb : 0 < b) :
  ( v in Ioc 0 b, (v ^ 2) ^ 1 * quadMomentPrimitive a (c * v))
  = c * quadPowerPrimitive a (c * b)
theorem integral_quadPowerPrimitive_div ... :
  ( x in Ioc 0 L, quadPowerPrimitive a x / x)
  = logPrimitive a L - quadPowerPrimitive a L
theorem threeDIntegralH001_eq ... : threeDIntegralH001 a b n
  = b / Real.sqrt n * (logPrimitive a (Real.sqrt n * b ^ 3)
  - quadPowerPrimitive a (Real.sqrt n * b ^ 3))
theorem threeDIntegralH001_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n => threeDIntegralH001 a b n) ~[atTop]
  fun n => b * fluctuation (1 / 2) a / 4 * (n ^ (-(1 / 2 : )) * Real.log n)
```

3 Proof strategy

The exact formula was first checked numerically (relative error 10^{-16} ; the ratio $\sqrt{n} Z/(b \log n)$ tends to $A/2 = S_{1/2}(a)/4$). The reduction is three applications of `setIntegral_congr_fun` with the substitution lemmas, each in the divide-and-substitute form of units 35–39 (valid at $u = 0$ because the primitives vanish there). The asymptotic follows from the L -uniform bounds: `logPrimitive_bounds` gives $H(L) = A \log L + O(1)$ with $\log L = \frac{1}{2} \log n + 3 \log b$, and $0 \leq K \leq A$, so $|Z - (bA/2) \log n/\sqrt{n}| \leq (b/\sqrt{n})(E + M + 3A|\log b| + A)$, which is $o(\log n/\sqrt{n})$ by `isLittle0_inv_sqrt_log`. The new global bound $|F_1(t)| \leq Et^2$ (all real t) makes the K -integrand bounded on $\mathbb{R}$, giving continuity of K via `continuous_primitive` and hence integrability of $K(x)/x$ on $(0, L]$ (bounded by $E/2$).

4 Roadblocks and resolutions

- `inv_mul_le_iff` produces $t^2 \cdot E$, not $E \cdot t^2$; a trailing `mul_comm` in the rewrite chain aligns it with the bound lemma. This was the only compile error.
- A docstring over 100 columns tripped the line-length gate after a clean build; the gate must run before committing, not after.

5 What was Mathlib, what was new

Mathlib: `norm_integral_le_of_norm_le_const`, `continuous_primitive`,
`integral_comp_mul_left`, `Integrable.mono'`, `IsBig0.of_bound`, `IsLittle0.isEquivalent`.
 New: the closed form $\int_0^L K/x = H - K$, the noncritical substitution $\int_0^b x^{-2} F_1(cx) = cK(cb)$, and the first mixed-multiplicity instance of `thm:TaylorTree`.

6 Lessons

In the mixed case the non-minimal coordinates do not need their own primitive hierarchy: after one noncritical substitution they enter as a bounded function subtracted from the critical primitive, and the critical coordinates alone determine the logarithmic power. This suggests the general (k, h) case will reduce to the all-equal machinery of unit 39 applied to a modified kernel.

7 Outcome

Units 35–40 now exhibit all three regimes of the multiplicity count in the simplest chart geometry: no logarithm (distinct), $(\log n)^{d-1}$ (all equal), and $\log n$ with a bounded correction (two of three equal). Next candidates: the weighted primitives F_γ for general (k, h) , or the next-order term $-Bn^{-1/2}$ in the logarithmic case.